

CONCEPT 7.5

Bulk transport across the plasma membrane occurs by exocytosis and endocytosis (pp. 139–141)

- In **exocytosis**, transport vesicles migrate to the plasma membrane, fuse with it, and release their contents. In **endocytosis**, molecules enter cells within vesicles that pinch inward from the plasma membrane. The three types of endocytosis are **phagocytosis**, **pinocytosis**, and **receptor-mediated endocytosis**.

? Which type of endocytosis involves the binding of specific substances in the extracellular fluid to membrane proteins? What does this type of transport enable a cell to do?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

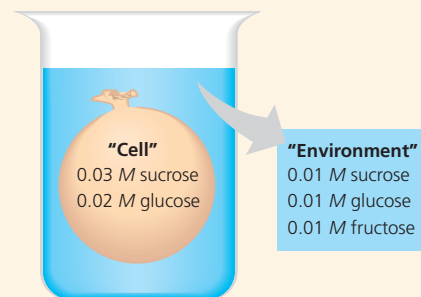
Levels 1-2: Remembering/Understanding

- In what way do the membranes of a eukaryotic cell vary?
 - Phospholipids are found only in certain membranes.
 - Certain proteins are unique to each membrane.
 - Only certain membranes of the cell are selectively permeable.
 - Only certain membranes are constructed from amphipathic molecules.
- According to the fluid mosaic model of membrane structure, proteins of the membrane are mostly
 - spread in a continuous layer over the inner and outer surfaces of the membrane.
 - confined to the hydrophobic interior of the membrane.
 - embedded in a lipid bilayer.
 - randomly oriented in the membrane, with no fixed inside-outside polarity.
- Which of the following factors would tend to increase membrane fluidity?
 - a greater proportion of unsaturated phospholipids
 - a greater proportion of saturated phospholipids
 - a lower temperature
 - a relatively high protein content in the membrane

Levels 3-4: Applying/Analyzing

- Which of the following processes includes all the others?
 - osmosis
 - diffusion of a solute across a membrane
 - passive transport
 - transport of an ion down its electrochemical gradient
- Based on Figure 7.19, which of these experimental treatments would increase the rate of sucrose transport into a plant cell?
 - decreasing extracellular sucrose concentration
 - decreasing extracellular pH
 - decreasing cytoplasmic pH
 - adding a substance that makes the membrane more permeable to hydrogen ions
- DRAW IT** An artificial “cell” consisting of an aqueous solution enclosed in a selectively permeable membrane is immersed in a beaker containing a different solution, the “environment,” as shown in the accompanying diagram. The membrane is permeable to water and to the simple sugars glucose and fructose but impermeable to the disaccharide sucrose.
 - Draw solid arrows to indicate the net movement of solutes into and/or out of the cell.
 - Is the solution outside the cell isotonic, hypotonic, or hypertonic?

- Draw a dashed arrow to show the net osmosis, if any.
- Will the artificial cell become more flaccid, more turgid, or stay the same?
- Eventually, will the two solutions have the same or different solute concentrations?



Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** *Paramecium* and other protists that live in hypotonic environments have cell membranes that limit water uptake, while those living in isotonic environments have membranes that are more permeable to water. Describe what water regulation adaptations might have evolved in protists in hypertonic habitats such as the Great Salt Lake and in habitats with changing salt concentration.
- SCIENTIFIC INQUIRY** An experiment is designed to study the mechanism of sucrose uptake by plant cells. Cells are immersed in a sucrose solution, and the pH of the solution is monitored. Samples of the cells are taken at intervals, and their sucrose concentration is measured. The pH is observed to decrease until it reaches a steady, slightly acidic level, and then sucrose uptake begins. (a) Evaluate these results and propose a hypothesis to explain them. (b) Predict what would happen if an inhibitor of ATP regeneration by the cell were added to the beaker once the pH was at a steady level. Explain.
- SCIENCE, TECHNOLOGY, AND SOCIETY** Extensive irrigation in arid regions causes salts to accumulate in the soil. (When water evaporates, salts that were dissolved in the water are left behind in the soil.) Based on what you learned about water balance in plant cells, explain why increased soil salinity (saltiness) might be harmful to crops.
- WRITE ABOUT A THEME: INTERACTIONS** A human pancreatic cell obtains O_2 —and necessary molecules such as glucose, amino acids, and cholesterol—from its environment, and it releases CO_2 as a waste product. In response to hormonal signals, the cell secretes digestive enzymes. It also regulates its ion concentrations by exchange with its environment. Based on what you have just learned about the structure and function of cellular membranes, write a short essay (100–150 words) to describe how such a cell accomplishes these interactions with its environment.
- SYNTHESIZE YOUR KNOWLEDGE**



In the supermarket, lettuce and other produce are often sprayed with water. Explain why this makes vegetables crisp.

For selected answers, see Appendix A.